

Weyl Spinor Signal in the Cosmological Higgs Collider

A direct two-Weyl calculation of the top-quark box

1 Two-Weyl setup and Feynman rules

1.1 Lagrangian, basis and mode functions

We use $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$, $\sigma^\mu = (1, \boldsymbol{\sigma})$, $\bar{\sigma}^\mu = (1, -\boldsymbol{\sigma})$, and $a(\tau) = -1/(H\tau)$, $\tau < 0$. The conformally rescaled fields have action

$$S = \int d\tau d^3x \left\{ i t^\dagger \bar{\sigma}^\mu \partial_\mu t + i t^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(\tau)(m_t + g_t \delta h)(t t^c + t^\dagger t^{c\dagger}) \right\}, \quad (1)$$

$$g_t = y_t/\sqrt{2}, \quad m_t = g_t h_0, \quad \mu = m_t/H, \quad N_c = 3. \quad (2)$$

The spinor epsilon conventions are chosen so that the displayed mass term is $-am_t \Psi_W^\dagger \beta_W \Psi_W$, with

$$\Psi_W = \begin{pmatrix} t_\alpha \\ t^{c\dagger\dot{\alpha}} \end{pmatrix}, \quad \beta_W = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \quad i\partial_\tau \Psi_W(\mathbf{p}) = \begin{pmatrix} -\boldsymbol{\sigma} \cdot \mathbf{p} & am_t \mathbf{1}_2 \\ am_t \mathbf{1}_2 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix} \Psi_W(\mathbf{p}). \quad (3)$$

This column is only a convenient way to keep the two Weyl equations together. Both entries have the same top charge. In particular no reality condition is imposed. We evaluate matrices in the orthonormal component basis

$$e_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1}_2 \\ \mathbf{1}_2 \end{pmatrix}, \quad e_- = \frac{1}{\sqrt{2}} \begin{pmatrix} -\mathbf{1}_2 \\ \mathbf{1}_2 \end{pmatrix}, \quad U_B = (e_+, e_-), \quad \beta = U_B^\dagger \beta_W U_B = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}. \quad (4)$$

This is a change of spinor-component basis, not the replacement of t, t^c by two Majorana fields. Charge remains proportional to the identity. In this basis,

$$H(\mathbf{p}, \tau) = \begin{pmatrix} M & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -M \end{pmatrix}, \quad M = -\mu/\tau. \quad (5)$$

For $\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \xi_h = h \xi_h$, $h = \pm 1$, use

$$\xi_+ = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad \xi_- = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}, \quad \sum_h \xi_h \xi_h^\dagger = \mathbf{1}_2. \quad (6)$$

The massless Higgs external contraction is the one used in the supplied notes:

$$G_a(k; \tau) = \frac{H^2}{2k^3} (1 - iak\tau) e^{iak\tau}, \quad a = \pm 1. \quad (7)$$

All external vectors are incoming, $\sum_i \mathbf{k}_i = 0$, and a prime removes the overall $(2\pi)^3 \delta^3(\sum_i \mathbf{k}_i)$. We consider exact de Sitter, constant m_t , zero chemical potential, and the leading collapsed expansion.

Write $x = -p\tau > 0$. A positive-frequency column has components $(F(x)\xi_h, hG(x)\xi_h)^T$, giving

$$-iF' = \frac{\mu}{x}F + G, \quad -iG' = F - \frac{\mu}{x}G, \quad (8)$$

$$F'' + \left[1 + \frac{\mu^2 + i\mu}{x^2}\right]F = 0, \quad G'' + \left[1 + \frac{\mu^2 - i\mu}{x^2}\right]G = 0. \quad (9)$$

The Bunch–Davies solutions, with a common irrelevant phase fixed, are

$$\boxed{F(x) = \frac{\sqrt{\pi x}}{2} e^{\pi\mu/2} H_{1/2-i\mu}^{(1)}(x), \quad G(x) = \frac{\sqrt{\pi x}}{2} e^{-\pi\mu/2} H_{1/2+i\mu}^{(1)}(x).} \quad (10)$$

Both tend to $-ie^{ix}/\sqrt{2}$ in the far past. The normalization is $|F|^2 + |G|^2 = 1$. At fixed Fourier momentum, a useful complete pair of mode matrices is

$$u(\mathbf{p}, \tau) = \begin{pmatrix} F\mathbf{1}_2 \\ G\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \end{pmatrix}, \quad v(\mathbf{p}, \tau) = \begin{pmatrix} G^*\mathbf{1}_2 \\ -F^*\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \end{pmatrix}. \quad (11)$$

The columns multiply independent annihilation and creation operators; the latter carry antiparticle label $-\mathbf{p}$. Thus $u^\dagger u = v^\dagger v = \mathbf{1}_2$, $u^\dagger v = 0$ and $uu^\dagger + vv^\dagger = \mathbf{1}_4$ at equal time. The original Weyl modes are $U_B u$ and $U_B v$. This also specifies their relative phases, which are essential for the soft cut. The mode construction is consistent with the charged-fermion momentum-space quantization of Ref. [1].

1.2 SK propagators

Define the contractions without a prefactor of i :

$$R^>(p; \tau, \tau') = u(\tau)u^\dagger(\tau'), \quad R^<(p; \tau, \tau') = -v(\tau)v^\dagger(\tau'). \quad (12)$$

The minus sign in $R^<$ is the fermionic exchange sign. Explicitly, with $\sigma_p = \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$,

$$R^> = \begin{pmatrix} F_1 F_2^* \mathbf{1}_2 & F_1 G_2^* \sigma_p \\ G_1 F_2^* \sigma_p & G_1 G_2^* \mathbf{1}_2 \end{pmatrix}, \quad R^< = \begin{pmatrix} -G_1^* G_2 \mathbf{1}_2 & G_1^* F_2 \sigma_p \\ F_1^* G_2 \sigma_p & -F_1^* F_2 \mathbf{1}_2 \end{pmatrix}. \quad (13)$$

The four contour contractions are

$$R_{-+} = R^>, \quad R_{+-} = R^<, \quad R_{++} = \theta(\tau - \tau')R^> + \theta(\tau' - \tau)R^<, \quad (14)$$

$$R_{--} = \theta(\tau' - \tau)R^> + \theta(\tau - \tau')R^<. \quad (15)$$

1.3 Feynman rules

The following rules refer to the conformally rescaled fermions in Eq. (2). SK contractions contain no extra factor of i . The branch sign at a vertex will be denoted by $s = \pm 1$ here to distinguish it from the scale factor $a(\tau)$.

1. **External Higgs line.** Attach $G_s(k; \tau)$ from Eq. (7) to a vertex on branch s . External Higgs fields are labelled before summing permutations.
2. **Yukawa vertex.** From $S_{\text{int}} = -g_t \int d\tau d^3x a(\tau) \delta h \Psi_W^\dagger \beta_W \Psi_W$, the vertex matrix is

$$-is g_t a(\tau) (\beta_W)_{AB} \delta_{cc'} = -is g_t a(\tau) \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}_{AB} \delta_{cc'}. \quad (16)$$

Here A labels the $\Psi_W^\dagger$ slot, B the Ψ_W slot, and c, c' are color indices. The two off-diagonal blocks are precisely the $\delta h t t^c$ and $\delta h t^\dagger t^{c\dagger}$ vertices with the index positions specified below. Each has coefficient $-is g_t a(\tau)$; there is no $1/2$ since the two Weyl fields are distinct. In the mode basis, replace β_W by $\beta = \text{diag}(\mathbf{1}_2, -\mathbf{1}_2)$. There is only one power of the scale factor at a vertex. The background mass is already included in the propagator, so it is not inserted again.

3. **Fermion line.** A line with field end (A, τ, s) and dagger end (B, τ', s') contributes

$$[R_{W,ss'}(\mathbf{p}; \tau, \tau')]_{AB} \delta_{cc'}. \quad (17)$$

where $R_W = U_B R U_B^\dagger$ and R is given in Eqs. (13)–(15). The block dictionary is

$$R_{W,ss'} = \begin{pmatrix} \langle t_\alpha t_\beta^\dagger \rangle_{ss'} & \langle t_\alpha t^{c\beta} \rangle_{ss'} \\ \langle t^{c\dagger\alpha} t_\beta^\dagger \rangle_{ss'} & \langle t^{c\dagger\alpha} t^{c\beta} \rangle_{ss'} \end{pmatrix}. \quad (18)$$

Thus the epsilon tensors needed to raise the t^c indices are already included; one must not add another epsilon tensor to the vertex matrix. This fixes the normal and anomalous Weyl contractions in one convention. The charge arrow runs from the dagger end to the field end, and the Fourier factor of this ordered line is $e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}')}$. If its endpoints are exchanged, use the Grassmann identity

$$\langle T_C \Psi_B^\dagger(\mathbf{p}, \tau) \Psi_A(-\mathbf{p}, \tau') \rangle_{ss'} = -[R_{s's}(-\mathbf{p}; \tau', \tau)]_{AB}. \quad (19)$$

In particular the minus sign of $R^<$ in Eq. (12) is part of the line rule, not an optional extra sign. Charge-forbidden contractions such as $\langle tt \rangle$ and $\langle t^c t^c \rangle$ vanish.

4. **Integrals and closed loop.** Integrate every vertex over $d\tau \in (-\infty, 0)$, impose spatial momentum conservation at each vertex, and integrate an independent loop momentum with $d^3q/(2\pi)^3$. Use the BD $i\epsilon$ prescription for the early-time endpoint, and sum all vertex branch labels. A closed fermion loop gives -1 , a spin trace, and a color trace $N_c = 3$. No spin average or additional factor of two is included. Each labelled directed box below has symmetry factor one.

Multiplying these rules gives Eq. (30); its overall sign is $(-1)_{\text{loop}}(-i)^4 = -1$, with the product of the four branch signs kept inside the SK sum.

1.4 Nonlocal part of the soft propagator

For $t = -\tau > 0$, the late-time expansion is

$$F(pt) = A(pt)^{i\mu} + \frac{iB}{1 - 2i\mu}(pt)^{1-i\mu} + O((pt)^2 \text{ times the branches}), \quad (20)$$

$$G(pt) = B(pt)^{-i\mu} + \frac{iA}{1 + 2i\mu}(pt)^{1+i\mu} + O((pt)^2 \text{ times the branches}), \quad (21)$$

$$A = -\frac{ie^{\pi\mu/2}2^{-1/2-i\mu}}{\sqrt{\pi}}\Gamma(\tfrac{1}{2} - i\mu), \quad B = -\frac{ie^{-\pi\mu/2}2^{-1/2+i\mu}}{\sqrt{\pi}}\Gamma(\tfrac{1}{2} + i\mu), \quad (22)$$

$$c = AB^* = \frac{2^{-1-2i\mu}}{\pi}\Gamma(\tfrac{1}{2} - i\mu)^2, \quad |c| = \frac{1}{2 \cosh \pi\mu}. \quad (23)$$

In particular,

$$R_{ab}^{\text{NL}}(\mathbf{p}; t, t') = c(p^2 tt')^{i\mu} \begin{pmatrix} 0 & \sigma_p \\ 0 & 0 \end{pmatrix} + c^*(p^2 tt')^{-i\mu} \begin{pmatrix} 0 & 0 \\ \sigma_p & 0 \end{pmatrix} + O(pt, pt'). \quad (24)$$

There is no SK label on this expression. For example, the upper-right $+i\mu$ term is $F(t)G^*(t')$ in $R^>$ and $G^*(t)F(t')$ in $R^<$; both equal $c(p^2 tt')^{i\mu}$. This equality also holds for their analytic momentum descendants. It is compatible with the absence of these powers in the state-independent anticommutator $R^> - R^<$. This property is independent of the choice between the two-Weyl and Majorana descriptions. In particular, the reverse Wightman contraction must use the reversed momentum in its negative-frequency helicity spinors. The two identities

$$h_s(\mathbf{k})h_s^\dagger(\mathbf{k}) = h_{-s}(-\mathbf{k})h_{-s}^\dagger(-\mathbf{k}), \quad (25)$$

$$[u_s(\tau_1)u_s^*(\tau_2)]_{\text{NL}} = -[v_s(\tau_2)v_s^*(\tau_1)]_{\text{NL}} \quad (26)$$

then give $[D_{-+} - D_{+-}]_{\text{NL}} = 0$ in the Majorana description too. The project reference PDF is v1 of Ref. [2], which used a different reverse-momentum expression and introduced a nonzero antisymmetric nonlocal sector. Version v2, Appendix D.3, Eqs. (237)–(240), explicitly includes $-\mathbf{k}$ and establishes the equality of the two nonlocal Wightman contractions. Thus the removal of this particular time-ordering obstruction is a correction to the propagator treatment, not a physical advantage of a Weyl basis. The fermionic minus sign was already present in v1; attributing the discrepancy solely to unsigned Wightman conventions would be incorrect. The old project Majorana box formulas therefore require revision before their final amplitudes can be compared with this calculation. No independent complete recomputation of that old box amplitude is asserted here.

For the first nonzero box signal we also need the first descendants. They are particularly transparent as the following explicit block product:

$$\beta R^{\text{NL}}(\mathbf{p}; t, t')\big|_+ = c(p^2 tt')^{i\mu} \begin{pmatrix} \mathbf{1}_2 \\ -\frac{it'\boldsymbol{\sigma} \cdot \mathbf{p}}{1 + 2i\mu} \end{pmatrix} \sigma_p \begin{pmatrix} -\frac{it'\boldsymbol{\sigma} \cdot \mathbf{p}}{1 + 2i\mu} & \mathbf{1}_2 \end{pmatrix} + \dots \quad (27)$$

The other branch is fixed by $R_-^{\text{NL}}(\mathbf{p}; t, t') = [R_+^{\text{NL}}(\mathbf{p}; t', t)]^\dagger$. The dots include second-order corrections to the leading entries. They multiply a vanishing zeroth-order Bose-symmetrized hard block and do not affect the first nonzero clock coefficient derived below.

2 Box diagrams and spin contraction

2.1 Diagram counting and four-point integrals

For four labelled external Higgs fields there are $(4 - 1)! = 6$ directed fermion cycles, or three undirected boxes with two orientations each. Each carries $-N_c$, and its symmetry factor is one. Expanding the Dirac determinant gives the same answer: the $1/4$ cancels cyclic rotations of the $4!$ labelled insertions. There is no extra Majorana factor, and color is counted once.

For $\pi = (i, j, k, l)$ define

$$p_{ij} = q + k_i, \quad p_{jk} = q + k_i + k_j, \quad p_{kl} = q - k_l, \quad p_{li} = q, \quad (28)$$

where this equation is a vector equation. The exact individual contribution is

$$\mathcal{G}_{ijkl} = -N_c g_t^4 \sum_{a_i, a_j, a_k, a_l = \pm} a_i a_j a_k a_l \int_{-\infty}^0 \prod_{v=i,j,k,l} [d\tau_v a(\tau_v) G_{a_v}(k_v; \tau_v)] \int \frac{d^3 q}{(2\pi)^3} \quad (29)$$

$$\times \text{tr}_4 [\beta R_{a_i a_j}(p_{ij}; \tau_i, \tau_j) \beta R_{a_j a_k}(p_{jk}; \tau_j, \tau_k) \beta R_{a_k a_l}(p_{kl}; \tau_k, \tau_l) \beta R_{a_l a_i}(q; \tau_l, \tau_i)]. \quad (30)$$

Here $(-i)^4 = 1$ from the four Yukawa vertices. The following table specifies Eq. (30) separately for all six diagrams, including all four line momenta. No unspecified permutation factor is implicit.

cycle	successive momenta	s -soft region	s -hard edges
1234	$q + k_1, q + k_s, q - k_4, q$	$q \sim k_s$	12, 34
1243	$q + k_1, q + k_s, q - k_3, q$	$q \sim k_s$	12, 43
1324	$q + k_1, q + k_1 + k_3, q - k_4, q$	none	—
1342	$q + k_1, q + k_1 + k_3, q - k_2, q$	$q + k_1 \sim k_s$	34, 21
1423	$q + k_1, q + k_1 + k_4, q - k_3, q$	none	—
1432	$q + k_1, q + k_1 + k_4, q - k_2, q$	$q + k_1 \sim k_s$	43, 21

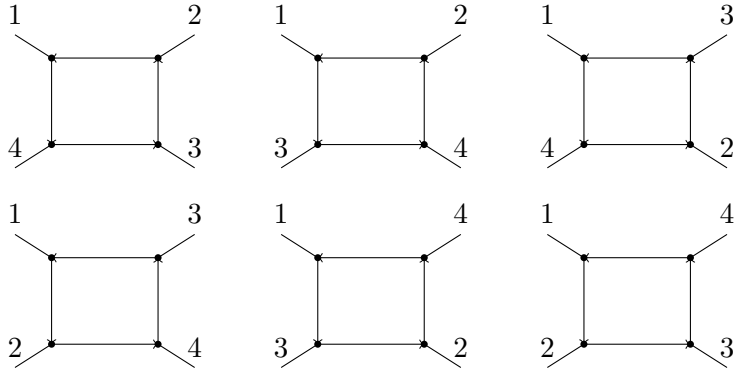


Figure 1: The six directed cycles in the order of the table. For the line R_{ij} the charge arrow points from j to i , opposite to the written matrix-cycle order. The Weyl propagator components and vertex chiralities are summed as in Eq. (31). The upper right and lower middle cycles have no two-soft-line cut in the s channel.

Here $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2$. Notice $p_{34} = q - k_4 = q + k_s + k_3$ in 1234; using $q + k_3$ at finite k_s would violate momentum conservation and spoil the first surviving signal. The t - and u -channel results follow by relabelling.

For an explicit two-Weyl component enumeration, expand the vertex matrices and propagators. The trace in each of the six expressions is the sum of the sixteen terms

$$\mathcal{S}_{ijkl}^w = \text{tr}_2[(\beta_W R_{W,ij})^{w_i w_j} (\beta_W R_{W,jk})^{w_j w_k} (\beta_W R_{W,kl})^{w_k w_l} (\beta_W R_{W,li})^{w_l w_i}], \quad w_v \in \{1, 2\}. \quad (31)$$

Each term's four-point function is Eq. (30) with its trace replaced by $\mathcal{S}^w$; its times, loop measure, SK signs and color factor are thus fully fixed. The component list is

word	successive blocks	word	successive blocks
1111	11,11,11,11	2111	21,11,11,12
1112	11,11,12,21	2112	21,11,12,22
1121	11,12,21,11	2121	21,12,21,12
1122	11,12,22,21	2122	21,12,22,22
1211	12,21,11,11	2211	22,21,11,12
1212	12,21,12,21	2212	22,21,12,22
1221	12,22,21,11	2221	22,22,21,12
1222	12,22,22,21	2222	22,22,22,22

These are 96 algebraic Weyl terms, not 96 distinct box topologies. The four s -cut cycles contain 64 of them. Keeping the matrix trace performs all these component contractions at once and avoids epsilon-index sign ambiguities.

2.2 Collapsed limit and Bose-symmetry cancellation

The collapsed kinematics are best organized symmetrically:

$$\mathbf{k}_1 = \mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_2 = -\mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad (32)$$

$$\mathbf{k}_3 = \mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_4 = -\mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad (33)$$

$$\boldsymbol{\ell} = \mathbf{q} + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s, \quad \mathbf{n}_L = \hat{\mathbf{k}}_L, \quad \mathbf{n}_R = \hat{\mathbf{k}}_R, \quad \mathbf{n}_s = \hat{\mathbf{k}}_s. \quad (34)$$

Write $k_L = |\mathbf{k}_L|$, $k_R = |\mathbf{k}_R|$; at the accuracy of the final answer $k_L = k_{12}/2$, $k_R = k_{34}/2$, where $k_{ij} = k_i + k_j$ is a sum of magnitudes.

For the + clock branch in 1234, the two projected hard matrices are

$$V_L = \begin{pmatrix} -\frac{it_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} & \mathbf{1}_2 \end{pmatrix} \beta R_{12}(\mathbf{k}_L + \boldsymbol{\ell}; t_1, t_2) \begin{pmatrix} \mathbf{1}_2 \\ -\frac{it_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \end{pmatrix}, \quad (35)$$

$$V_R = \begin{pmatrix} -\frac{it_3 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} & \mathbf{1}_2 \end{pmatrix} \beta R_{34}(\mathbf{k}_R + \boldsymbol{\ell}; t_3, t_4) \begin{pmatrix} \mathbf{1}_2 \\ -\frac{it_4 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \end{pmatrix}. \quad (36)$$

After the Weyl-component sum the entire spin contraction is

$$c^2 (qK)^{2i\mu} \prod_v t_v^{i\mu} \text{tr}_2[V_L \sigma_K V_R \sigma_q]. \quad (37)$$

Thus, already before integration,

$$\text{tr}[(\boldsymbol{\sigma} \cdot \mathbf{A}) \sigma_K (\boldsymbol{\sigma} \cdot \mathbf{B}) \sigma_q] = 2[(\mathbf{A} \cdot \hat{K})(\mathbf{B} \cdot \hat{q}) + (\mathbf{A} \cdot \hat{q})(\mathbf{B} \cdot \hat{K}) - (\mathbf{A} \cdot \mathbf{B})(\hat{K} \cdot \hat{q})]. \quad (38)$$

At zeroth order $V_L = -R_{21}(k_L) \propto \boldsymbol{\sigma} \cdot \mathbf{n}_L$ and $V_R \propto \boldsymbol{\sigma} \cdot \mathbf{n}_R$. Their spin trace is odd under either $\mathbf{n}_L \rightarrow -\mathbf{n}_L$ or $\mathbf{n}_R \rightarrow -\mathbf{n}_R$. The sum of the four cut cycles is precisely the independent exchange of the two left and the two right Higgs fields. Therefore

$$\boxed{\mathcal{G}_{s,\text{clock}}^{\text{box}} \Big|_{k_s^{3 \pm 4i\mu}} = 0.} \quad (39)$$

The cancellation follows before any scalar integration: every zeroth-order term contains one factor of each hard direction and is odd under either Higgs-pair exchange. We must therefore keep the first momentum correction at both ends of the cut.

3 Scalar integrals for the hard lines and the soft bubble

3.1 Left and right hard subgraphs

We now calculate the first order of each Bose-symmetrized hard block. Write $x = k_L t_1$, $y = k_L t_2$ and $r = p/k_L$. We need only two SK-summed scalar seeds, corresponding to the 21 and 11 blocks of R :

$$\mathcal{H}_{IJ}(\alpha, \beta; r) = \sum_{a,b=\pm} ab \int_0^\infty dx dy x^{\alpha-1} y^{\beta-1} (1 + iax)(1 + iby) e^{-iax-iby} [R_{IJ}(r; x, y)]_{ab}^{\text{radial}}, \quad (40)$$

$IJ = 21, 11.$

“Radial” means the coefficient of σ_p in R_{21} or of $\mathbf{1}_2$ in R_{11} ; their mode arguments are rx, ry . Hence these are scalar integrals, with the spin structure already removed. Their evaluated series is given immediately below in Sec. 3.2.

The useful exact reversal identities are

$$R_{21;ab}(t_1, t_2) = R_{21;ba}(t_2, t_1), \quad R_{11;ab}(t_1, t_2) = -R_{22;ba}(t_2, t_1). \quad (41)$$

For one left ordering, expanding the endpoint projectors gives

$$V_L = -R_{21} - \frac{it_1}{1 + 2i\mu} \boldsymbol{\sigma} \cdot \mathbf{q} R_{11} + \frac{it_2}{1 + 2i\mu} R_{22} \boldsymbol{\sigma} \cdot \mathbf{K} + O(q^2, k_s^2). \quad (42)$$

All entries here are unapproximated hard SK contractions before expansion of their momentum $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$. The other ordering has $\mathbf{p} = -\mathbf{k}_L + \boldsymbol{\ell}$. The derivative of the hard line and the descendants of both soft endpoints must both be kept. Eq. (41) shows that the first-order difference between the two external magnitudes gives zero in the sum: the integrated R_{21} block is symmetric in those energies. Also $\int t_2 R_{22} = -\int t_1 R_{11}$ at equal external magnitudes.

Strip $H^2/(4k_1^3 k_2^3)$ from the left two external lines and vertices. Its integrated projected matrix, summed over its two Higgs orderings, is

$$\mathcal{V}_L = -2k_L^{-1-2i\mu} \boldsymbol{\sigma} \cdot [\mathcal{P}_1(\mu) \boldsymbol{\ell} + \mathcal{P}_2(\mu) \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell})], \quad (43)$$

$$\mathcal{P}_1(\mu) = \mathcal{H}_{21}(i\mu, i\mu; 1) + \frac{2i}{1 + 2i\mu} \mathcal{H}_{11}(1 + i\mu, i\mu; 1), \quad (44)$$

$$\mathcal{P}_2(\mu) = (\partial_r - 1) \mathcal{H}_{21}(i\mu, i\mu; r) \Big|_{r=1}. \quad (45)$$

The two coefficients $\mathcal{P}_1, \mathcal{P}_2$ multiply the independent spin structures $\boldsymbol{\sigma} \cdot \boldsymbol{\ell}$ and $(\boldsymbol{\sigma} \cdot \mathbf{n}_L)(\mathbf{n}_L \cdot \boldsymbol{\ell})$. The $\mathcal{H}_{11}$ term comes from the soft endpoint descendants; the r derivative comes from the hard propagator's momentum dependence. The right result is the same with $(k_L, \mathbf{n}_L) \rightarrow (k_R, \mathbf{n}_R)$ and external prefactor $H^2/(4k_3^3 k_4^3)$. In particular the four-cycle sum is the product of these two already symmetrized blocks; no further factor of two or four is allowed. Formula (45) retains all terms at this order. The absence of the constant term in each block makes the signal a spatial P -wave fermion-pair effect. It adds two powers of k_s to the bubble scaling.

3.2 Evaluation of the scalar hard seeds

We give an explicit representation so the answer is a function of mass, rather than a definition by two unevaluated time integrations. The construction uses only scalar Hankel functions and ordered scalar Laplace integrals. Its analytic continuation is part of the definition, just as for scalar double-folded seeds.

A mode $Z \in \{F, G, \bar{F}, \bar{G}\}$ has the expansion

$$Z(rx) = \sum_{\eta=\pm 1} \sum_{n=0}^{\infty} z_{\eta n}(rx)^{\lambda_{\eta n}}, \quad \lambda_{\eta n} = \frac{1}{2} + \eta\nu + 2n. \quad (46)$$

For F use $(N, \nu) = (\sqrt{\pi}e^{\pi\mu/2}/2, 1/2 - i\mu)$; for G use $(\sqrt{\pi}e^{-\pi\mu/2}/2, 1/2 + i\mu)$. Their coefficients are

$$z_{-n} = \frac{N}{i \sin \pi\nu} \frac{(-1)^n 2^{\nu-2n}}{n! \Gamma(n+1-\nu)}, \quad (47)$$

$$z_{+n} = -\frac{N e^{-i\pi\nu}}{i \sin \pi\nu} \frac{(-1)^n 2^{-\nu-2n}}{n! \Gamma(n+1+\nu)}. \quad (48)$$

For a barred mode conjugate both coefficients and exponents; μ is real. These formulas follow immediately from $H_\nu^{(1)} = (J_{-\nu} - e^{-i\pi\nu} J_\nu)/(i \sin \pi\nu)$.

For a monomial $x^{A-1}y^{B-1}$ define two evaluated kernels,

$$K_{\text{all}}(A, B; s, t) = \frac{\Gamma(A)\Gamma(B)}{s^A t^B}, \quad (49)$$

$$K_{<}(A, B; s, t) = \frac{\Gamma(A+B)}{A t^{A+B}} {}_2F_1(A+B, A; A+1; -s/t), \quad (50)$$

$$K_{>}(A, B; s, t) = K_{<}(B, A; t, s). \quad (51)$$

The $<$ kernel integrates the region $x < y$, which is $\tau_1 > \tau_2$. It follows by setting $x = zy$, $0 < z < 1$, and doing the y Gamma integral.

For two specified modes Z, W define the *explicit double series*

$$\mathcal{L}_{ab, \Omega}^{ZW}(\alpha, \beta; r) = \text{FP}_{\delta \rightarrow 0} \text{AC}_{\epsilon \rightarrow 0^+} \sum_{\eta, n; \zeta, m} z_{\eta n} w_{\zeta m} r^{\lambda_{\eta n} + \lambda_{\zeta m}} \sum_{j, l=0}^1 (ia)^j (ib)^l \quad (52)$$

$$\times K_{\Omega}(\alpha + \delta + \lambda_{\eta n} + j, \beta + \delta + \lambda_{\zeta m} + l; \epsilon + ia, \epsilon + ib), \quad \Omega \in \{\text{all}, <, >\}. \quad (53)$$

Here AC means analytic continuation of the *summed* function from large positive Laplace damping (where termwise integration converges) to the displayed BD boundary value. One must not insert

$\epsilon = 0$ into a raw power-series truncation outside its convergence domain. The endpoint regulator δ is common to all terms. In the following combinations both regulator operations are performed *after the full SK sum*. This specifies the finite part unambiguously and preserves cancellations. For numerical use one may continue in the Laplace variables in steps, or use the equivalent convergent scalar Hankel contours; no additional frequency/PV integrations are involved.

The complete SK-summed off-diagonal and diagonal hard seeds are respectively

$$\mathcal{H}_{21}(\alpha, \beta; r) = \mathcal{L}_{++,<}^{G\bar{F}} + \mathcal{L}_{++,>}^{\bar{F}G} + \mathcal{L}_{--,>}^{G\bar{F}} + \mathcal{L}_{--,<}^{\bar{F}G} - \mathcal{L}_{+-,all}^{\bar{F}G} - \mathcal{L}_{-+,all}^{G\bar{F}}, \quad (54)$$

$$\mathcal{H}_{11}(\alpha, \beta; r) = \mathcal{L}_{++,<}^{F\bar{F}} - \mathcal{L}_{++,>}^{\bar{G}G} + \mathcal{L}_{--,>}^{F\bar{F}} - \mathcal{L}_{--,<}^{\bar{G}G} + \mathcal{L}_{+-,all}^{\bar{G}G} - \mathcal{L}_{-+,all}^{F\bar{F}}. \quad (55)$$

All terms on a line share $(\alpha, \beta; r)$. Inserting these two series into Eq. (45) evaluates $\mathcal{P}_1(\mu), \mathcal{P}_2(\mu)$. For $\mathcal{P}_2$, multiply each term of the $\mathcal{H}_{21}(i\mu, i\mu; 1)$ series by $\lambda_{\eta n} + \lambda_{\zeta m} - 1$; no numerical differentiation is needed. The derivative holds the external Higgs energies fixed. The regulator limits are taken after the complete SK sum. These are Gamma- $_2F_1$ expressions for the two hard coefficients, rather than unevaluated time integrals; a finite Gamma-only reduction is not assumed.

3.3 Soft bubble integral

The spin trace of Eq. (38) leaves a polynomial numerator divided by qK . To display it, write the two vectors appearing in the hard blocks as

$$\mathbf{v}_L = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell}), \quad \mathbf{v}_R = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_R (\mathbf{n}_R \cdot \boldsymbol{\ell}). \quad (56)$$

Since $\mathbf{K} = \boldsymbol{\ell} + \mathbf{k}_s/2$ and $\mathbf{q} = \boldsymbol{\ell} - \mathbf{k}_s/2$, the scalar loop integral after the spin contraction is

$$2 \int \frac{d^3 q}{(2\pi)^3} (qK)^{2i\mu-1} \left[2(\mathbf{v}_L \cdot \boldsymbol{\ell})(\mathbf{v}_R \cdot \boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L \cdot \mathbf{k}_s)(\mathbf{v}_R \cdot \mathbf{k}_s) - (\mathbf{v}_L \cdot \mathbf{v}_R)(\ell^2 - k_s^2/4) \right]. \quad (57)$$

Its mass dimension is $5 + 4i\mu$. We now evaluate it by scalar convolutions, following the same procedure as for the vector box. The only required scalar formula is

$$F_0(A, B) = \frac{\Gamma(A+B-\frac{3}{2})\Gamma(\frac{3}{2}-A)\Gamma(\frac{3}{2}-B)}{\Gamma(A)\Gamma(B)\Gamma(3-A-B)}, \quad (58)$$

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-A} (K^2)^{-B} = \frac{k_s^{3-2A-2B}}{(4\pi)^{3/2}} F_0(A, B). \quad (59)$$

The right-hand side is the analytic continuation of the nonanalytic part; UV-dependent polynomial contact terms are excluded.

For the reduction only, measure q, K, ℓ in units of k_s , put $u = q^2$, $v = K^2$, and set $\nu = 1/2 - i\mu$. Then

$$\ell^2 = (u + v)/2 - 1/4, \quad \ell_{\parallel} \equiv \boldsymbol{\ell} \cdot \mathbf{n}_s = (v - u)/2, \quad \ell_{\perp}^2 = \ell^2 - \ell_{\parallel}^2. \quad (60)$$

Azimuthal averaging replaces two transverse components by $\ell_{\perp}^2 \delta_{ij}^{\perp}/2$ and four by $\ell_{\perp}^4 (\delta_{ij}^{\perp} \delta_{kl}^{\perp} + \delta_{ik}^{\perp} \delta_{jl}^{\perp} + \delta_{il}^{\perp} \delta_{jk}^{\perp})/8$, where $\delta_{ij}^{\perp} = \delta_{ij} - n_{si} n_{sj}$. The numerator in Eq. (57) consequently becomes a polynomial in u, v and the three external angles. Each monomial is integrated by

$$u^m v^n \longmapsto \frac{F_0(\nu - m, \nu - n)}{(4\pi)^{3/2}}, \quad (61)$$

$$\frac{F_0(\nu - m, \nu - n)}{F_0(\nu, \nu)} = \frac{(3/2 - \nu)_m (3/2 - \nu)_n (1 - \nu)_m (1 - \nu)_n}{(5/2 - 2\nu)_{m+n} (3 - 2\nu)_{m+n}}. \quad (62)$$

Only nonnegative integers m, n with $m + n \leq 4$ occur. Thus the bubble is a finite Gamma sum, and all its terms reduce to the same $F_0(\nu, \nu)$ times rational functions. For example the coefficient of $\mathcal{P}_1^2$ in the square bracket of Eq. (57) is $\ell^4 + \ell^2/4 - \ell_{\parallel}^2/2$, whose integral is

$$\frac{F_0(\nu, \nu)}{(4\pi)^{3/2}} \frac{3(\nu - 1)(2\nu - 3)}{8(\nu - 2)(4\nu - 7)(4\nu - 5)}. \quad (63)$$

The $\mathcal{P}_1\mathcal{P}_2$ and $\mathcal{P}_2^2$ terms are evaluated by the same two steps. Their fully reduced coefficients are displayed directly in Eq. (71), without introducing separate symbols for each tensor moment.

4 Nonlocal signal

4.1 Oscillatory signal and angular coefficients

Choose the angle convention

$$\mathbf{n}_L = (\sin \theta_1, 0, \cos \theta_1), \quad \mathbf{n}_R = (\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \quad \mathbf{n}_s = (0, 0, 1). \quad (64)$$

The leading nonzero oscillatory term, summed over the four s -cut boxes, is

$$\begin{aligned} \langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle' |_{s, \text{clock}} = & - \frac{N_c g_t^4 H^4 k_s^5}{16 k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R} \\ & \times \left[\left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\mu} \mathcal{C}(\mu; \theta_1, \theta_3, \phi_3) + \text{c.c.} \right]. \end{aligned} \quad (65)$$

The coefficient is given directly in the angular basis of the vector-box note:

$$\begin{aligned} \mathcal{C} = & A_1 + A_2(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3 \cos^2 \theta_1 \cos^2 \theta_3 \\ & + B_1 \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 + D_1 \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \end{aligned} \quad (66)$$

All five coefficients share one soft-bubble normalization, including the two soft-mode coefficients c^2 , the factor 4 from the two Bose-summed hard blocks, the factor 2 from the Pauli trace, and the phase from the chosen momentum ratio:

$$\mathcal{B}(\mu) = \frac{2^{1+4i\mu} \Gamma(\frac{1}{2} - i\mu)^4 F_0(\nu, \nu)}{32\pi^2 (4\pi)^{3/2} (\nu - 3)(\nu - 2)(4\nu - 7)(4\nu - 5)}, \quad \nu = \frac{1}{2} - i\mu. \quad (67)$$

Combining the bubble with the two hard coefficients of Eq. (45) gives the final mass dependence:

$$A_1 = \mathcal{B}(\nu - 1)(2\nu - 3) [4(\nu - 3)(3\mathcal{P}_1^2 + 2\mathcal{P}_1\mathcal{P}_2) + (2\nu - 5)\mathcal{P}_2^2], \quad (68)$$

$$A_2 = -\mathcal{B}\mathcal{P}_2^2(\nu - 1)(2\nu - 3)^2, \quad A_3 = -3A_2, \quad (69)$$

$$B_1 = -2\mathcal{B}\mathcal{P}_2^2(\nu - 1)(\nu - 2)(2\nu - 3), \quad (70)$$

$$D_1 = \mathcal{B}\mathcal{P}_2^2(\nu - 2)(2\nu - 5)(2\nu - 3). \quad (71)$$

Thus the answer requires only two hard functions $\mathcal{P}_1, \mathcal{P}_2$ and one explicit bubble factor $\mathcal{B}$. The coefficients are complex and include the clock phase. The full answer is even under either Higgs-pair exchange and under parity. Its engineering dimension is -5 , as required for the primed four-point function. The powers of H follow from four rescaled Weyl Yukawa vertices and four external Higgs contractions; no extra factor of H or a second color/orientation multiplicity is implicit. Higher soft powers and analytic contact terms are not displayed.

4.2 Nonoscillatory contribution

Taking opposite soft branches gives a nonoscillatory k_s^3 term. The hard blocks are R_{11} and $-R_{22}$; Eq. (41) makes their integrated values equal to the real seed $\mathcal{H}_{11}(-i\mu, +i\mu; 1)$. The scalar bubble in this case is

$$\int \frac{d^3q}{(2\pi)^3} (q^2)^{-i\mu} (K^2)^{i\mu} \hat{q} \cdot \hat{K} = \frac{k_s^3}{(4\pi)^{3/2}} \frac{2\sqrt{\pi}}{3} \mu(1 + \mu^2) \coth(\pi\mu). \quad (72)$$

After both soft branches and all four s -cut boxes are summed, the result is

$$\langle \delta h^4 \rangle' \Big|_{s, \text{mixed}} = \frac{N_c g_t^4 H^4 k_s^3}{16 \prod_i k_i^3} \frac{2\mu(1 + \mu^2)}{3\pi \sinh(2\pi\mu)} \left[\mathcal{H}_{11}(-i\mu, +i\mu; 1) \right]^2. \quad (73)$$

It is independent of angles at this order. Equation (73) is the leading nonoscillatory nonanalytic term, whereas Eq. (65) is the leading mass-dependent clock signal. At $\mu = 0$ the separated branches coalesce; the limit must be taken after combining the relevant terms.

Checks and scope. The accompanying scripts check the mode equations, normalization, signed SK contractions, soft descendants, spin traces, scalar Gamma reductions, and hard momentum expansion. A numerical scan of the analytically continued hard seeds has not been performed. The calculation assumes constant mass in exact de Sitter, a massless external Higgs and zero chemical potential. The supplied gauge-boson note provides the scalar-factorization workflow and angular convention; the top-quark note provides g_t, μ, G_a . The loop factorization follows Ref. [3]. No Majorana-field decomposition is used.

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